

cuOpt benchmark: GS Retail routing

T2 · Appendix A1–A2 · CVRPTW · 537–586 orders

Internal benchmark reported by Omelet. Results below are transcribed from the exact final-pitch appendix, not newly rerun tests. They support discussion of experimental performance; they do not establish shipped NVIDIA SDK adoption. Original run logs and named test-owner confirmation are not attached; independent verification is pending.

Metric	Route distance (km) and per-instance gap to the best result among compared solvers; 10- and 60-second budgets.
Problem	Three GS Retail instances; A3 replenishment scenario; 537–586 orders.
Setup	Full-results appendix specifies one RTX A6000, August 2026, drive-distance objective and 55 km/h. Summary appendix also mentions RTX 4090 / A6000; exact per-run hardware mapping is not supplied. Repeated runs and re-validation are reported.
Test date / owner	August 2026 as reported for the GS Retail scenario. Exact dates, number of repeats and responsible engineer/team: not yet verified.

METHOD	10 S: KM	10 S: GAP	60 S: KM	60 S: GAP
NVIDIA cuOpt	2,684	+1.09%	2,610	+0.83%
Omelet RouteOpt	2,676	+1.37%	2,622	+1.70%
RouteOpt, ALNS configuration	2,984	+12.22%	2,766	+7.16%
OR-Tools	3,000	+12.79%	2,815	+9.50%
LKH-3	No feasible solution	at 1,830 s	≈30× the 60 s budget	—

How to interpret the results

Gap uses a moving, per-instance best-across-solvers reference. Compare distance across time budgets, not the gap percentages as if they shared a fixed global optimum. Aggregated km and mean per-instance gap may rank differently; raw per-instance records are needed to reconstruct the averages.

The appendix reports 2/3 cuOpt wins at both budgets and run-to-run variation: cuOpt ± 1.24 / ± 0.50 percentage points and RouteOpt-ALNS ± 0.87 / ± 1.27 percentage points (10 / 60 seconds). The deck describes the frontier difference as within noise. A formal significance/equivalence test and comparable variation for primary RouteOpt are not supplied; no statistical-tie conclusion is independently verified here.

These are solver distance results. They do not independently prove the separate 20%+ realized delivery-cost reduction claim. Only aggregate results are published; customer orders and routes are not included.

Source and reproducibility

[Open appendix A2: full results](#)

Needed: instance identifiers or approved anonymized instances, per-run distance/feasibility records, aggregation formula, CPU details, exact solver versions/configurations and named test owner.